

15-(MathAA/12_HL_Summer_2022_Q5)-BinomialTheorem

[Maximum mark: 7]

Consider the binomial expansion $(x + 1)^7 = x^7 + ax^6 + bx^5 + 35x^4 + \dots + 1$ where $x \neq 0$ and $a, b \in \mathbb{Z}$.

(a) Show that $b = 21$.

[2]

The third term in the expansion is the mean of the second term and the fourth term in the expansion.

(b) Find the possible values of x .

[5]



1-(MAT/12_HL_Summer_2012_Q1)-Remainder&FactorTheorem

The same remainder is found when $2x^3 + kx^2 + 6x + 32$ and $x^4 - 6x^2 - k^2x + 9$ are divided by $x + 1$. Find the possible values of k .

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



2-(MAT/10_HL_Winter_2013_Q1)-Remainder&FactorTheorem

[Maximum mark: 5]

The cubic polynomial $3x^3 + px^2 + qx - 2$ has a factor $(x+2)$ and leaves a remainder 4 when divided by $(x+1)$. Find the value of p and the value of q .

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



3-(MAT/11_HL_Summer_2014_Q1)-Remainder&FactorTheorem

[Maximum mark: 5]

When the polynomial $3x^3 + ax + b$ is divided by $(x-2)$, the remainder is 2, and when divided by $(x+1)$, it is 5. Find the value of a and the value of b .

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

exam-mate.co



4-(MAT/10_HL_Winter_2017_Q3)-Remainder&FactorTheorem

[Maximum mark: 6]

Consider the polynomial $q(x) = 3x^3 - 11x^2 + kx + 8$.

- (a) Given that $q(x)$ has a factor $(x - 4)$, find the value of k . [3]
- (b) Hence or otherwise, factorize $q(x)$ as a product of linear factors. [3]



5-(MAT/11_HL_Summer_2018_Q1)-Remainder&FactorTheorem

[Maximum mark: 5]

Let $f(x) = x^4 + px^3 + qx + 5$ where p, q are constants.

The remainder when $f(x)$ is divided by $(x + 1)$ is 7, and the remainder when $f(x)$ is divided by $(x - 2)$ is 1. Find the value of p and the value of q .



6-(MAT/11_HL_Summer_2019_Q10)-Remainder&FactorTheorem,Probability

The function $p(x)$ is defined by $p(x) = x^3 - 3x^2 + 8x - 24$ where $x \in \mathbb{R}$.

(a) Find the remainder when $p(x)$ is divided by

(i) $(x - 2)$

(ii) $(x - 3)$.

[3]

(b) Prove that $p(x)$ has only one real zero.

[4]

(c) Write down the transformation that will transform the graph of $y = p(x)$ onto the graph of $y = 8x^3 - 12x^2 + 16x - 24$.

[2]

The random variable X follows a Poisson distribution with a mean of λ and $6P(X = 3) = 3P(X = 2) - 2P(X = 1) + 3P(X = 0)$.

(d) Find the value of λ .

[6]



7-(MathAA/10_HL_Winter_2022_Q11)-Permutation-Combination,Remainder&FactorTheorem

[Maximum mark: 16]

Consider a three-digit code abc , where each of a , b and c is assigned one of the values 1, 2, 3, 4 or 5.

(a) Find the total number of possible codes

- (i) assuming that each value can be repeated (for example, 121 or 444);
- (ii) assuming that no value is repeated. [4]

Let $P(x) = x^3 + ax^2 + bx + c$, where each of a , b and c is assigned one of the values 1, 2, 3, 4 or 5. Assume that no value is repeated.

Consider the case where $P(x)$ has a factor of $(x^2 + 3x + 2)$.

- (b)
- (i) Find an expression for b in terms of a .
 - (ii) Hence show that the only way to assign the values is $a = 4$, $b = 5$ and $c = 2$.
 - (iii) Express $P(x)$ as a product of linear factors.
 - (iv) Hence or otherwise, sketch the graph of $y = P(x)$, clearly showing the coordinates of any intercepts with the axes. [12]

